KB-009-Safety-Protocols

KB-009 — Safety Escalation Protocols

Category: Safety — ALWAYS ACTIVE
Call Volume: Low — but highest priority of any issue
Authority: Any agent can and must escalate safety concerns immediately
Primary Actions: Immediate disconnect instruction, emergency services referral, safety dispatch

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Critical Rule

Safety escalations override all other call flows. If a driver mentions any of the safety triggers listed below — mid-troubleshooting, mid-billing dispute, at any point — stop the current flow and go to the appropriate safety script immediately.

Safety Trigger Recognition

Escalate immediately if the driver mentions any of the following:

Trigger phrase / description	Category	Response
“Sparks,” “sparking,” “arc,” “flash of light”	Electrical arc	Section A
“Burning smell,” “smoke,” “it smells like something’s burning”	Thermal event / fire risk	Section A
“Fire,” “flames,” “it’s on fire”	Active fire	Section A — call emergency services
“I got a shock,” “it shocked me,” “electric shock,” “tingling”	Electrical contact	Section B

Trigger phrase / description	Category	Response
“The connector is melted,” “the cable is melted,” “it’s discolored and hot”	Thermal damage	Section C
“My car is making a strange noise from the battery,” “my car smells like sulfur”	Battery thermal event (vehicle)	Section D
“The charger fell over,” “it looks structurally damaged,” “the cabinet is open”	Physical safety hazard	Section E
“There’s water in the charging port,” “flooding,” “standing water near the charger”	Water / electrical hazard	Section F
“My car battery swelled,” “puffed up,” “the battery looks wrong”	Battery deformation	Section D

Section A — Electrical Arc, Smoke, or Fire

This is the highest-priority scenario.

Script (read verbatim if possible)

> “I need you to stop what you’re doing right now. Please do not touch the connector or the charger. If the cable is plugged in and safe to reach without getting near the sparking area, unplug it — but only if it’s safe. Otherwise, step away from the charger and your vehicle immediately. Get at least 30 feet away.”

If there are visible flames or active smoke:

> “Please call 911 immediately — do not wait. Tell them there is an electrical fire at an EV charging station. Then step away from the vehicle. Do not attempt to put out an EV battery fire with a conventional extinguisher — those fires require special suppression. Get to safety and let the fire department handle it.”

Agent actions (immediately after driver is safe)

1. **Flag the station as SAFETY EMERGENCY** in the backend — immediately taken offline, no remote resets, no other drivers can start sessions
2. **Dispatch a safety-priority technician** — not a standard tech ticket. This is a priority safety inspection.
3. **Log the incident** with: station ID, date/time, driver name and contact, description of what was observed, whether driver was injured
4. **Notify the on-call supervisor** — all safety incidents involving arc, fire, or smoke require supervisor notification within 15 minutes
5. **Do not reset or unlock anything** on this station until a technician has physically inspected it

If driver reports they've already left the area

> "That was the right call. Is everyone okay? Are you safe right now?"

Confirm safety, then proceed with logging. The station is already flagged. Advise the driver that a technician will inspect the station and they may be contacted for more details if an incident report is needed.

Section B — Electric Shock

Electric shock from an EV charger can range from a mild static-like sensation to a serious injury. Treat all shock reports seriously.

Immediate script

> "I'm very sorry to hear that. First — are you feeling okay right now? Are you in pain, feeling dizzy, or having any unusual sensations?"

If driver reports significant pain, difficulty breathing, numbness, muscle spasms, or any signs of serious injury:

> "Please call 911 immediately. Electric shock can have internal effects that aren't immediately obvious. Please get medical attention right now — I can stay on the line with you. Do not drive if you feel unwell."

If driver reports only a mild, brief sensation (similar to static):

> "I understand — are you feeling okay now? Any tingling or numbness?"

If they feel fine: > "I'm glad you're okay. Please don't touch the connector or the charger again — I'm flagging this station for a safety inspection right now."

Agent actions

1. **Flag station as SAFETY HOLD** in the backend immediately
2. **Log incident:** station ID, time, driver name and contact, description of the shock (mild/severe, where on body, duration), any reported injury
3. **Dispatch technician** for electrical safety inspection — ground fault testing required
4. **Notify supervisor** — all shock incidents require supervisor notification
5. **Document driver's stated condition** — if they decline emergency services but later develop symptoms, this log is important

Common causes of charger-related shock sensations

- **Ground fault** at the charger (the most serious — triggers E005/GFCI error; charger should self-disable)
- **Static electricity** — not from the charger itself, but from the vehicle or environment; harmless but startling
- **Damaged cable insulation** — a serious safety fault; station must be taken offline immediately
- **Faulty vehicle inlet** — if the vehicle's inlet has insulation damage, the shock originates at the vehicle; advise vehicle inspection

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Section C — Melted or Thermally Damaged Connector / Cable

If a driver describes a connector that appears melted, discolored, deformed, or extremely hot to the touch:

Script

> "Please do not touch the connector again — it may still be at unsafe temperatures, and the damage could indicate an ongoing electrical issue. Step away from the charger."

If the connector is still in the vehicle:

> "Do not try to remove it. A technician will need to do that safely. Is the session showing as ended on the screen?"

If session is ended and cable is not in the vehicle: advise the driver not to pick up the connector.

Agent actions

1. **Flag station as SAFETY HOLD** — immediately taken offline
2. **Dispatch priority technician** with a note: "Thermally damaged connector — do not restart until inspection"

3. **Advise driver to inspect their vehicle's charging inlet** for signs of heat damage — if they see discoloration, they should not charge the vehicle again until the inlet is inspected by their dealer
4. **Log:** station ID, connector description, driver contact info, vehicle make/model
5. **Notify supervisor**

Why connector overheating happens

- **Loose connection** — a slightly unseated connector creates resistance, which generates heat; repeated over many sessions, this can cause progressive thermal damage
- **Pin wear / corrosion** — worn or corroded connector pins create resistance at contact points
- **Overcurrent fault** — charger delivers more current than the cable rating; hardware fault
- **Damaged cable insulation** — exposes conductors; risk of arc and fire

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Section D — Vehicle Battery Thermal Event

Lithium-ion batteries can undergo "thermal runaway" — a self-sustaining exothermic reaction that generates intense heat and fire. This is rare but extremely dangerous. EV battery fires burn very hot, can reignite after appearing extinguished, and require massive water application to cool (not conventional extinguishers).

Trigger signs (driver describes)

- Swelling or deformation of the vehicle body near the battery
- Sulfur or "rotten egg" smell coming from the vehicle
- Smoke coming from under the vehicle
- Battery warning light plus unusual heat
- Crackling, hissing, or popping sounds from under the vehicle floor

Script

> "Please get away from the vehicle immediately — at least 100 feet if possible. Do not get back in the car. Call 911 right now and tell them you have a potential EV battery fire. Tell them the vehicle make, model, and color."

> "Are you able to get away from the vehicle safely?"

Confirm the driver is moving to safety. Stay on the line if possible.

> "The fire department will need a lot of water to cool a lithium battery fire — they're trained for this. Please don't attempt to put it out yourself. I'm flagging this as an emergency on our end."

Agent actions

1. **This is a 911 emergency** — confirm driver has called or is calling
2. **Flag station as emergency shutdown** — do not allow any use until fire department and technician have cleared it
3. **Immediately notify supervisor** — this is a critical incident requiring executive notification per most network incident response plans
4. **Do not remotely reset, unlock, or take any remote action** on the station until cleared by emergency responders
5. **Document:** time, station ID, vehicle details, driver contact, nature of the event

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Section E — Physical Damage to Charger Structure

If a driver reports the charger has fallen over, the cabinet is open or damaged, or there is physical damage exposing internal components:

Script

> "Please step away from the charger — don't touch it. Exposed internal components can be at high voltage even when the station appears off."

> "Is the cable still plugged into a vehicle?"

If cable is in vehicle: > "Don't try to remove the cable. I'm dispatching a technician right now."

If cable is not in vehicle: > "Leave the cable where it is and stay away from the unit. I'm flagging it now."

Agent actions

1. **Flag station as SAFETY HOLD**
2. **Dispatch technician** — note: physical damage, possible exposed internals
3. **Log:** description of damage, whether cable is still in a vehicle, station ID, driver contact

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Section F — Water / Flooding Near Charger

EV charging stations are rated for outdoor use and weather exposure (typically IP54 or higher), but standing water, flooding, or direct water entry into the unit is a safety concern.

Script

> "If there's standing water near or under the charger, please don't approach it. Water and electrical equipment can be a dangerous combination even with safety ratings."

If the charger is clearly submerged or flooded: > "Do not approach the charger at all. I'm flagging this station for emergency inspection."

If the water is minimal (puddle from rain, not flooding): > "Modern chargers are designed to handle rain and wet conditions, but I'm going to flag this one for inspection to be safe. Is the charger showing any error messages?"

Post-Incident Documentation

All safety incidents require a complete log before the call ends. Collect:

Field	Required
Station ID	Always
Date and time of incident	Always
Driver name	Always
Driver contact (phone, email)	Always
Vehicle make, model, year	Always
Description of what was observed	Always
Whether emergency services were contacted	Always
Driver's reported physical condition	Always
Whether driver accepted or declined medical attention	Always
Remote actions taken (if any)	Always
Technician dispatch confirmed	Always
Supervisor notified	Always

Supervisor Notification Thresholds

Incident type	Notification timeline
Electric shock — any severity	Within 15 minutes
Arc, sparking, fire	Within 15 minutes
Vehicle battery thermal event	Immediately
Thermally damaged connector	Within 30 minutes
Physical damage with exposure risk	Within 30 minutes
Driver reported injury (any)	Within 15 minutes

Agent Reminders

- **Safety overrides everything.** Do not finish a billing script, a session reset, or any other flow if a safety trigger is present.
- **Confirm the driver is safe before logging.** The log can wait. The driver cannot.
- **Never reset a station that has a safety flag.** Even if a driver requests it.
- **Do not minimize a shock report.** What feels like "just a tingle" can mask internal injury. Always recommend medical attention for any reported shock.
- **EV battery fires are not like other fires.** A vehicle fire must be handled by trained fire department — never advise a driver to use a fire extinguisher on a battery fire.
- **You have the authority to dispatch and flag.** You do not need supervisor approval to take a station offline for safety. Act first, notify after.